

Real Estate Insights Is tokenization of real estate ready for lift off in APAC?

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Abstract

Purpose – Blockchain and distributed ledger technologies are set to disrupt the real estate sector in all areas: ownership, sale, management and investment. Tokenization moves physical real estate to the digital space and could result in substantial cost savings in the pre- and post-tokenization process. This article discusses whether real estate as an asset class is ready for digitalization in the Asia-Pacific (APAC) region.

Design/methodology/approach – Globally, the APAC region has the highest digital adaptation/adoption rates. Regulators in the region are also moving fast to clarify their stance on digital assets. This article adopts a holistic view, from trends, regulations, and technology, to discuss the benefits and challenges of digitalizing real estate in APAC.

Findings – Real estate tokenization is a nascent market but platforms like BrickX, KASA, ADDX, and Minterest have successfully launched real estate tokens in Australia, South Korea, and Singapore, respectively. Tokenization may prove to be a viable funding source for those relatively poorly capitalized financial markets in the APAC region.

Practical implications – This paper discusses the current regulatory and business contexts in relation to the pace of tokenization of real estate in APAC. Opportunities and difficulties are outlined in a concise manner to facilitate more discussion in this area.

Originality/value – Existing reports and research articles tend to focus on the western markets. This article provides a new perspective on tokenization, specifically in the APAC context.

Keywords Tokenization, Blockchain, Distributed ledger, APAC

Paper type Viewpoint

Introduction

Generally, the difficulties of investing in real estate are: indivisibility, large capital investment, illiquid, lack of information especially for developing or closed markets, long settlement time, long search process for the right property, high transaction costs (lawyers, consultants, valuers, taxes, fees, etc.) and high administrative costs for funds (dividend distribution, updates to the register of owners, etc.) Consequently, retail investors are usually excluded from investing in real estate, either due to capital constraints or access to the projects. Problems related to real estate investing also deter savvy investors from allocating a larger portion of their funds to real estate. According to Baum (2020), in 2019, institutional investors only allocated 10% of their funds to real estate though portfolio theory supports a more optimal asset allocation of 30–60%.

The advent of securitization mitigated the divisibility issue – stakeholders could now buy shares of a real estate investment trust (REIT) and enjoy the revenues originating from the physical asset. However, governance problems related to sponsor–satellite relationships (first-right of refusal for sponsors' properties), nonalignment of interest between shareholders and asset managers (non-yield-accretive acquisitions that increase the asset base) are some of the highlight problems with REITs. The rapid evolution of technology has solved many “old world” problems in the present financial system: high cost of money transfers, high proportions of the population that are “unbanked” or “underserved” by the banking system, high barrier to investing in physical real estate



for the retail investors, etc. Can the application of blockchain technology to tokenize real estate really be the solution to “democratize” real estate investment and open access to the common person on the street?

This opinion piece aims to provide an overview of the benefits and challenges of tokenizing physical real estate using blockchain technologies, and argues that this technological disruption can provide the foundation to build platforms that can democratize real estate for retail investors and consequently, provide direct access for small and medium developers/asset owners to connect with retail investors without going through intermediaries.

Generally, tokenization of real estate refers to the issuance of real estate digital assets on a blockchain platform using distributed ledger technologies (DLTs). The tokens will be encoded with the identity of the token holder and specify the ownership of economic benefits derived from the underlying asset (ownership, revenue-sharing, profit-sharing, etc.) To illustrate the use of tokenization, let us describe a use case from Thailand – the SiriHub Token.

Use case – SiriHub Token [1]

On September 21, 2021 Thailand launched SiriHub Token, the first regulated real estate-backed, public initial coin offering (ICO) housed on the Tezos blockchain network. The offering consists of only one office asset (Siri Campus Office Building valued at THB 2.4 billion/USD 73 million) as the underlying security. The office asset has a long-term lease commitment of 12 years to a single tenant Sansiri PCL. The token was designed for a term of four years and revenues from both the rent and sale of asset will be distributed to the token holders.

The SiriHub Token was offered in two tranches (A, B) to cater for investors with different risk appetite. During the investment period, token holders are entitled to the rents collected (distributed quarterly; capped at 4.5% p.a. and 8% p.a., respectively for Tranches A and B). At the end of the term, the asset will be auctioned with a minimum reserve of net proceeds THB 1.6 billion. The issuers expect the land value to appreciate at least 10% in the future.

If the sale is successful, token holders take a share of the sales proceeds (Tranche A takes first bite and profit share up to THB 1.6 billion of the sales proceeds; subsequently, the excess net sale proceeds go to Tranche B holders). If the minimum reserve is not met, tranche A token holders hold the right to vote on whether to proceed with the sale. The minimum investment for this ICO was priced at THB 10 (US\$0.30). Subscription for the tokens was via an online app (XSpring) available 24/7. The SiriHub Token was approved by Thailand’s Securities and Exchange Commission (SEC) and on October 14, 2021 began trading at the digital exchange managed by ERX Company Limited.

This use case perfectly illustrates the benefits of tokenization over the traditional securitization method:

- (1) *Fractional ownership*: ownership of the economic benefits is fractionalized to very small denominations via tranches (rental and price appreciation sharing).
- (2) *Customization*: economic benefits are customized to investors with different risk appetites.
- (3) *Small capitalization offering*: small projects or single asset securitization is currently not viable under public listing requirements.
- (4) *Absence of financial intermediaries*: cuts cost of floating a small-scale project on the public financial market.

Use of blockchain technologies

Two critical components of blockchain technologies – smart contracts and distributed ledger technologies – help to enable the cost and process efficiencies needed for tokenization to work. Without these disruptive technologies, fractional ownership is not realizable due to prohibitive administrative, transactional and reporting costs.

The benefits of tokenization have been extolled by both academics and practitioners [2]. In a nutshell, benefits accrue to launching the projects in the primary market and managing the project pre- and post-tokenization:

- (1) *Fractionalization*: Further divisible from one unit share to fraction of a share, and automated and managed via a digital register of members on the blockchain.
- (2) *Flexibility*: This ties in with fractionalization. Investors could now customize their exposure by directly buying into the assets or revenue streams.
- (3) *Operational efficiency*: Smart contracts programmed to automate pre-issuance matters like compliance checks, comply with know-your-customer (KYC) and anti-money laundering (AML) regulations, and post-issuance matters like dividend distribution and customizable fee structures.
- (4) *Contemporaneous settlement time*: Transactions could be settled almost instantly rather than T+ as in the traditional financial transactions.
- (5) *Information permanence and transparency*: All information is recorded and captured via DLTs, while security keys and digital addresses ensure data anonymity. Moving forward, DLTs could also house all information about the property (land purchase records, liens, title registry, expenses, revenues) which would greatly reduce compliance and verification costs.
- (6) *Liquidity*: With single-asset or specific revenue streams being tokenized, investors could now readily turn their illiquid assets that generate economic benefits into something transferable and traded.
- (7) *Access to new investors*: Blockchain technologies allow global access and also a new class of investors (tech savvy young investors) to enter the market in a language and platform they are comfortable in, 24/7.

Challenges to tokenization

Baum (2020) clearly stated the challenges of tokenizing single real estate assets: (1) sketchy evidence of investor demand for digital tokenization of single real estate assets; (2) regulations governing the fractionalization of ownership of assets and (3) market participants need to be comfortable in using digital platforms. Compared to tokenizing real estate assets, tokenizing debt securities originated from real estate may be the way forward, as financial markets can tap on existing systems to implement it. Baum (2020) also stated that the opinions were formed based on the European market. The question here is will these key challenges carry over to APAC markets?

Prior to SiriHub, since December 2020, Kasa (another platform in South Korea) had successfully issued digital asset-backed securities (DABS) backed by commercial real estate located in Seoul. To date, Kasa has raised a total of US\$19.26 million for the three office buildings (Yeoksam Korea Technology Center, Yeoksam Londonville and Seocho Gwell Tower). The digital tokens were sold to 12,751 retail investors. In addition, the demand for such alternative investments can be inferred from the active trades in Kasa's secondary exchange – the average daily traded value of the tokens is way in excess of US\$1 million per day.

Referring to the SiriHub Tokens, 240 million tokens were fully subscribed by more than 6,000 investors, raising THB 2.4 billion. On Oct 14, 2021, within the first 12 h of open trading, the SiriHub Token had traded over US\$450,000 in value [3]. Real Capital Analysis and Colliers estimated the total addressable market for security tokens in APAC to be US\$268 billion by 2025 [4]. At present, there is a lack of data to confirm retail investor demand for real estate tokens. However, anecdotally, several platforms like BrickX (Australia), Kasa (South Korea), ADDX and Minterest (Singapore) are seeing strong interest from both institutional and retail investors.

Baum (2020) critically pointed out that the issue with tokenization of real estate resides with the laws governing fractional ownership of land and property. However, as pointed in the same paper, this could be easily solved via the REIT-issuance structure with the establishment of a special purpose investment vehicle to house legal ownership and carry out the necessary asset maintenance activities. Similar issues are present in the APAC region. Regulations have yet to evolve to allow fractional ownership of land; thus, to date, the structuring of real estate tokens still follow the REIT structure.

A related issue with regulation lies with recognition or regulation of real estate as digital assets. Drawing parallel comparison with the turmoil currently experienced in the cryptocurrency markets, regulation or recognition by governments of the tokens issued is a major factor that directly affects the establishment and growth of tokenization in the real estate sector. Respondents to a survey [5] by the Asia Securities Industry and Financial Markets Association (ASIFMA) highlighted regulatory and legal uncertainty as the main challenges to tokenized securities.

Previous attempts at garnering alternative funding for real estate on a digital platform include crowdfunding (arguably started in 2003 in the USA by Prodigy Network) and ICOs in 2016–2017. In APAC, Fundplaces was one of the earliest recognized tokenization platforms that was launched in July 2017 in Singapore [6]. According to an MIT (2019) research paper, real estate crowdfunding failed to take off mainly due to adverse selection of properties and multi-year lock in periods. ICOs were the first crowdfunding application on the blockchain but failed to take off because almost all the ICOs were speculative “raise capital first and then find the assets” type; the coin sponsors lack the expertise to carry out the acquisitions and management of assets post fundraising. These highly speculative ICOs involved significant risks leading to public warnings and enforcement actions by regulators. The question is what makes tokenization different.

By now, regulators have shown a clear position that real estate tokens are recognized as “securities” subjected to stringent listing processes similar to an initial public offering (IPO). Following the fiasco of “The DAO” offering in 2017, the United States SEC clarified their position on what investment offerings falls under the definition of “securities”. The Howey Test is widely used to determine whether tokens or coin offerings are “securities”. In the APAC region, key financial centers like Japan, Singapore and Hong Kong have been increasing regulatory clarity on the security token market. In March 2019, The Securities and Futures Commission of Hong Kong published a new licensing regime for digital asset service providers. In May 2020, the Monetary Authority of Singapore laid down guidelines for the tokenization of assets. Also effective in May 2020, amendments to the Japanese Financial Instruments and Exchange Act of 2019 clarified that security token offerings (STOs) and ICOs are regulated under the Act. Singapore and Hong Kong have also launched sandbox programs in relation to offers or issues of digital tokens. These regulation clarities by financial regulators provide structure and legality to the offerings of security tokens, a situation that differs from previous ICOs and crowdfunding offerings.

Finally, we address the point about comfort level of investors using the digital platform. Pre-coronavirus disease 2019 (COVID-19), APAC had the fastest digital adoption or adaptation rates globally. The COVID-19 pandemic accelerated adoption of digital services and

technologies for consumers, businesses and even governments. The consulting firm, McKinsey, in a survey conducted in 2020 found [7] that during the COVID-19 lockdown, digital technologies played an important role in providing universal access and digital inclusion leading to an unprecedented increased reliance on digital infrastructure and services. The greatest impact of accelerated adoption of digital technologies was in the APAC region: customer adoption was accelerated by four years, while business digitization for products and/or services by 10+ years. Practically, the ubiquity of using apps for everyday needs like grocery shopping, online shopping, online banking, social media, etc. has intricately integrated the use of digital services into consumers' mindsets and behaviors. It is only a matter of time that investing in financial products using apps becomes the norm. In addition, the growing number of investors in cryptocurrency and non-fungible tokens (NFTs) will also bring about a bigger pool of investors who are able, ready and willing to invest in real estate tokens.

Further digital challenges

One of the critical factors that challenges the growth of this sector lies in the tradability of the tokens. Tokens can be traded over the blockchain, but the issue lies with the establishment of a centralized secondary market for STOs. At present, unlike stocks, there is no centralized marketplace to facilitate the trading of the tokens. Akin to the situation in the cryptocurrency market, where cryptocurrencies are traded across many different exchanges at different prices, real estate tokens can only be traded in the digital exchange that they are listed. Referring back to SiriHub Token, it can only be traded on the digital exchange managed by ERX Company Limited. If the digital exchange does not have the market depth and sufficient number of participants for trading, the liquidity problem resurfaces.

In addition, the lack of a centralized marketplace could lead to excessive search costs and transaction costs for investors in finding the right exchanges to trade their various real estate tokens. Information on the price and volatility of the tokens will have to be separately collected from the diverse platforms, hence, affecting the market efficiency of tokens. The use of separate blockchain networks raised the issue of siloed data and interoperability among different networks. However, compared to the current situation where retail investors have no options to invest in real estate (except through funds), the nascent digital market for real estate tokens still has a lot of room to grow in solving these teething problems.

Currently, all real estate tokens launched in the APAC region do not use cryptocurrency on their platforms. Investors pay fiat money in exchange for the digital tokens. In the ideal end state of blockchain application, trading of the real estate tokens would be paid via cryptocurrencies executed on smart contracts. The lack of a credible digital currency to streamline the payments rails results in lost opportunities to further decrease transaction and administration costs in real estate transactions.

The increasing interest in the establishment of central bank digital currencies (CBDCs) is a promising development. According to the Atlantic Council [8] that tracks the development of CBDCs, 83 countries are exploring the adoption of CBDCs. Much has been developed in the wholesale CBDC (central banks or major financial institutions for wholesale transactions) area, but few countries are fully implementing retail CBDC (for public use). Here in APAC, wholesale CBDC has been tested by governments in Singapore, Thailand and Hong Kong [9]. In Oct 2020, Cambodia launched Project Bakong which is a payment system infrastructure based on blockchain technology. Though Singapore has yet to announce concrete initiatives for a retail CBDC, on Oct 2020, a stable coin (ticker: XSGD) that exchanges 1:1 with the Singapore Dollar was launched by Xfers. In Feb 2021, China started trials of the digital yuan in major locations (Shenzhen, Suzhou, Shanghai, etc.) These developments are encouraging, as digital currencies would further aid in lowering administration and transaction in the post-tokenization processes.

The way forward for APAC real estate?

Real estate tokenization is a nascent market that shows great potential in “democratizing” real estate as an asset class to investors. Tokenization allows slicing and dicing of economic benefits (rental, price appreciation, etc.), participation in the various stages of real estate development (land acquisition, development, construction, etc.), and fractionalization of ownership units. These benefits allow investors to better customize their investment according to their investment constraints (capital, risk, etc.) and geographic exposure to real estate. Tokenization also reduces cost and time in the pre- and post-tokenization process. Increasingly, a rising number of users familiar with blockchain platforms through NFTs and cryptos will create a larger pool of investors for real estate tokens.

The blockchain technology is also gaining widespread acceptance due to better technology governance, security and reliability. Governments are also spearheading large-scale initiatives to transform business processes which will reinforce market acceptance of digital investment platforms. The most daunting challenges remain the “old world” issues pertaining to contractual relationship and legislative treatment of real estate ownership. In many ways, in jurisdictions like Singapore, intermediary professionals in the real estate transaction supply chain, that is, legal, tax and audit, are coming to grips with processes involving security tokenization. Regulatory bodies are adapting their regulatory requirements to cater to the digital world, with new laws enacted and existing laws reviewed to cater to these new digital instruments. While these developments take time, it is clear that the necessary rails are being built to enable tokenization to speed along at the appropriate time.

The relatively poorly capitalized financial system in the APAC region provides a good breeding ground for greater demand from developers and asset owners for capital raising via alternative platforms. In APAC, most real estate markets are dominated by large conglomerations for mega-projects which are well funded from traditional banks whilst many smaller developers struggle to meet even basic funding needs. This disproportionately drives up the financing costs of development in smaller projects. Tokenization of single asset or projects provide an excellent alternate channel for these developers to connect directly with small funders, build a verifiable track record and reduce overall cost of development. The success of Kasa in Korea and SiriHub in Thailand has given the real estate tokenization story a real buzz, and we are likely to see more such platforms in the coming years.

In conclusion, we believe that the concurrent growth of user base, changing regulatory stance and stabilization of the blockchain technology provide a good breeding ground in APAC to grow tokenization platforms in the near future.

Notes

1. Factual information gathered from various sources (accessed Oct 15, 2021): <https://www.kaohoon.com/news/479482>; <https://www.bangkokpost.com/business/2190863/token-creates-phenomenon-with-public-ico>; and <https://www.nationthailand.com/pr-news/business/40006399>
2. Information summarized on Oct 18, 2021 from: <https://assets.kpmg/content/dam/kpmg/cn/pdf/en/2020/04/real-estate-tokenization.pdf>; and <https://mitcre.mit.edu/wp-content/uploads/2019/11/Tokenized-Security-Commercial-Real-Estate2.pdf>
3. <https://xtz.news/adoption/tezos-based-sto-sirehub-token-issued-for-73-million-usd-by-xspring-and-traded-over-usd450k-in-the-first-12-hours-on-erx/> (accessed Oct 20, 2021)
4. <https://www2.deloitte.com/cn/en/pages/audit/articles/pr-real-estate-sto-whitepaper-1st-paper-report.html> (accessed Oct 20, 2021)
5. <https://www.asifma.org/wp-content/uploads/2021/06/tokenised-securities-in-apac-a-state-of-play-june-2021-final.pdf> (accessed Oct 20, 2021)

6. <https://www.businesstimes.com.sg/real-estate/singapore-firms-dip-toes-into-blockchain-tech-for-property>
7. <https://www.mckinsey.com/business-functions/strategy-and-corporate-finance/our-insights/how-covid-19-has-pushed-companies-over-the-technology-tipping-point-and-transformed-business-forever#> (accessed Oct 21, 2021)
8. <https://www.atlanticcouncil.org/cbdctracker/> (accessed Oct 21, 2021)
9. <https://www.theasianbanker.com/updates-and-articles/r3-ghosh-asia-pacific-countries-lead-the-charge-towards-cbdcs> (accessed Oct 21, 2021)

References

- Baum, A. (2020), *Tokenisation – The Future of Real Estate Investment?*, University of Oxford Research, available at: <https://www.sbs.ox.ac.uk/sites/default/files/2020-01/tokenisation.pdf> (accessed 15 October 2021).
- MIT Digital Currency Initiative (2019), *Tokenized Securities and Commercial Real Estate*, available at: <https://mitcre.mit.edu/wp-content/uploads/2019/11/Tokenized-Security-Commercial-Real-Estate2.pdf> (accessed 15 October 2021).

Further reading

- University of Oxford Research (2020), *PropTech 2020: The Future of Real Estate*, available at: <https://www.sbs.ox.ac.uk/sites/default/files/2020-02/proptech2020.pdf> (accessed 15 October 2021).

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